# 6.3.2 Backward error analysis of dot product: general case

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Week 06 due Feb 21, 2025 11:05 IST

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**Let's move on to the general case of dot product, where we take the dot product of**

two vectors of size n each. And if you write it out, then the exact result would

require the given multiplications and additions. And we're going to assume that

0:01 / 6:48

2.0x

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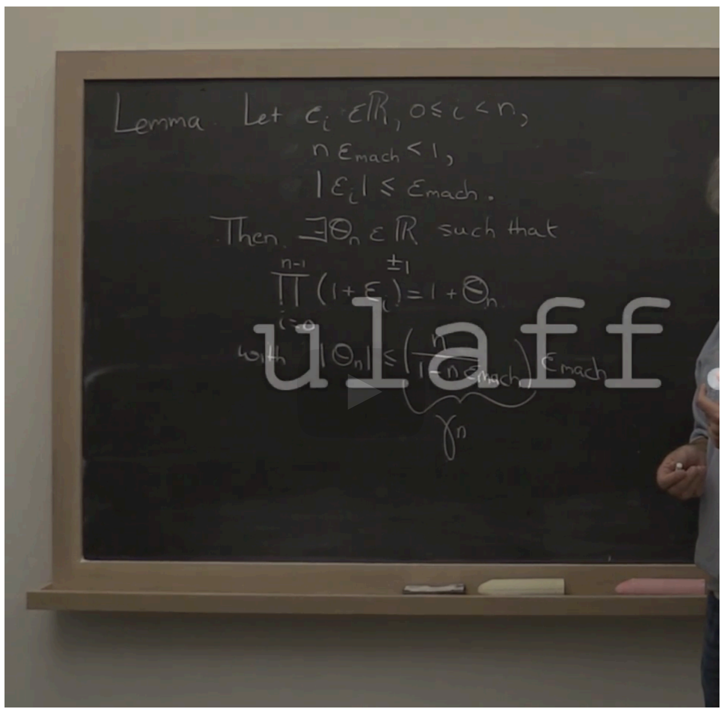


1 plus theta_n

is equal to the product of all of those factors. Okay? Moreover you can prove

that if each of the epsilons in magnitude is less than machine epsilon,

and notice that the absolute value of theta n is then less than something

times the machine epsilon where that something

is a function of n. The proof of this --well, you can kind of look at that and say,

"Well probably I need to prove that by

3:44 / 3:48

2.0x

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With this, we can take this really ugly expression if we had previously and

write it much more concisely as the computed value when you do your dot

products with vectors of size n is given by the exact dot product, except that now

we throw these factors in of 1 plus theta_n for the first term, 1 plus

theta_n for the

2:11 / 2:11

2.0x

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## Homework 6.3.2.2

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Prove Lemma 6.3.2.5.

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